

CLIMATE RESILIENCE AND HYDROMETEOROLOGICAL DISASTER MANAGEMENT

LIGTAS

LoRa-Integrated Grassroots Typhoon Alert System

Flood warnings that work when the internet doesn't, and disaster cash that arrives in days, not weeks.

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GitHub: github.com/Raizellrou/ligtas · Live demo: pwa-five-mocha.vercel.app

Three things fail at once. A fourth is never listed.

The warning channel dies

Towers and internet go down, so cloud early warning is unreachable when it is needed. Free cell broadcast under RA 10639 dies with the towers.

National data is not local action

A resident cannot turn an orange rainfall warning over the province into "leave now, go to the school, avoid the riverside road."

No affordable local sensing

Commercial river gauges are far outside a barangay budget, so most barangays have no real-time water data at all.

The records get disputed

After a disaster, whether warnings were issued and when is argued over, and the records sit with the parties being questioned.

THE IDEA

Five cheap radio boxes. No phone company.

One at the river with water sensors, two or three on rooftops as relays, one at the barangay hall. They talk by radio, kilometres apart, so they survive the storm.

Solar powered

LoRa, kilometres of range

Siren and hub WiFi at the hall

How a signed alert travels

1 • Threshold crossed

Water passes a set level at the river node.

2 • Packet signed

Hazard, tier, affected puroks, time and sequence number, signed with Ed25519.

3 • Hub verifies

Signature and sequence are checked offline. Forged or replayed packets are rejected.

4 • Siren and WiFi

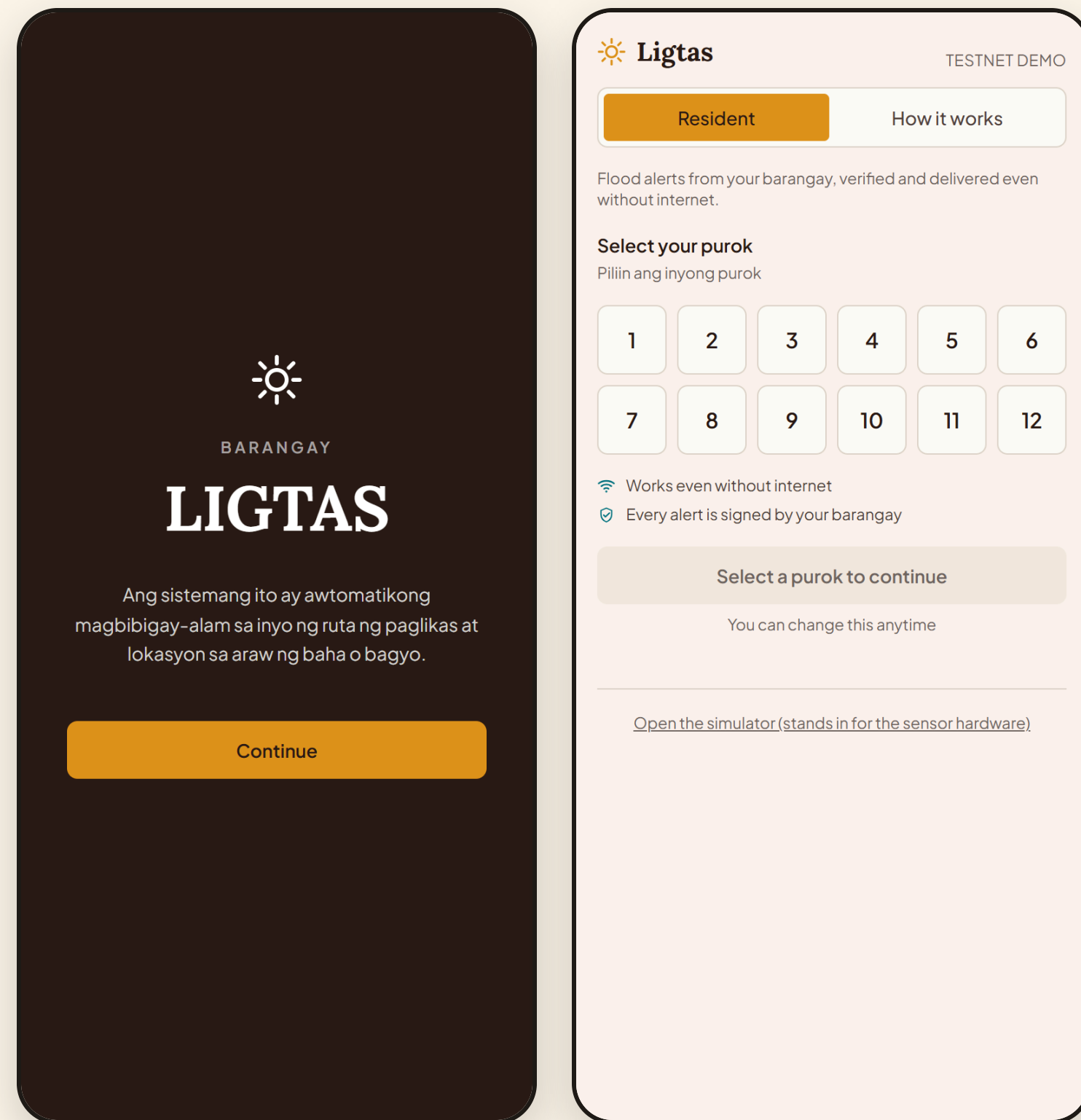
The hall sounds a siren and serves a page on its own network.

5 • Resident sees

An instruction for their own purok and the route to the nearest center.

6 • Chain keeps receipt

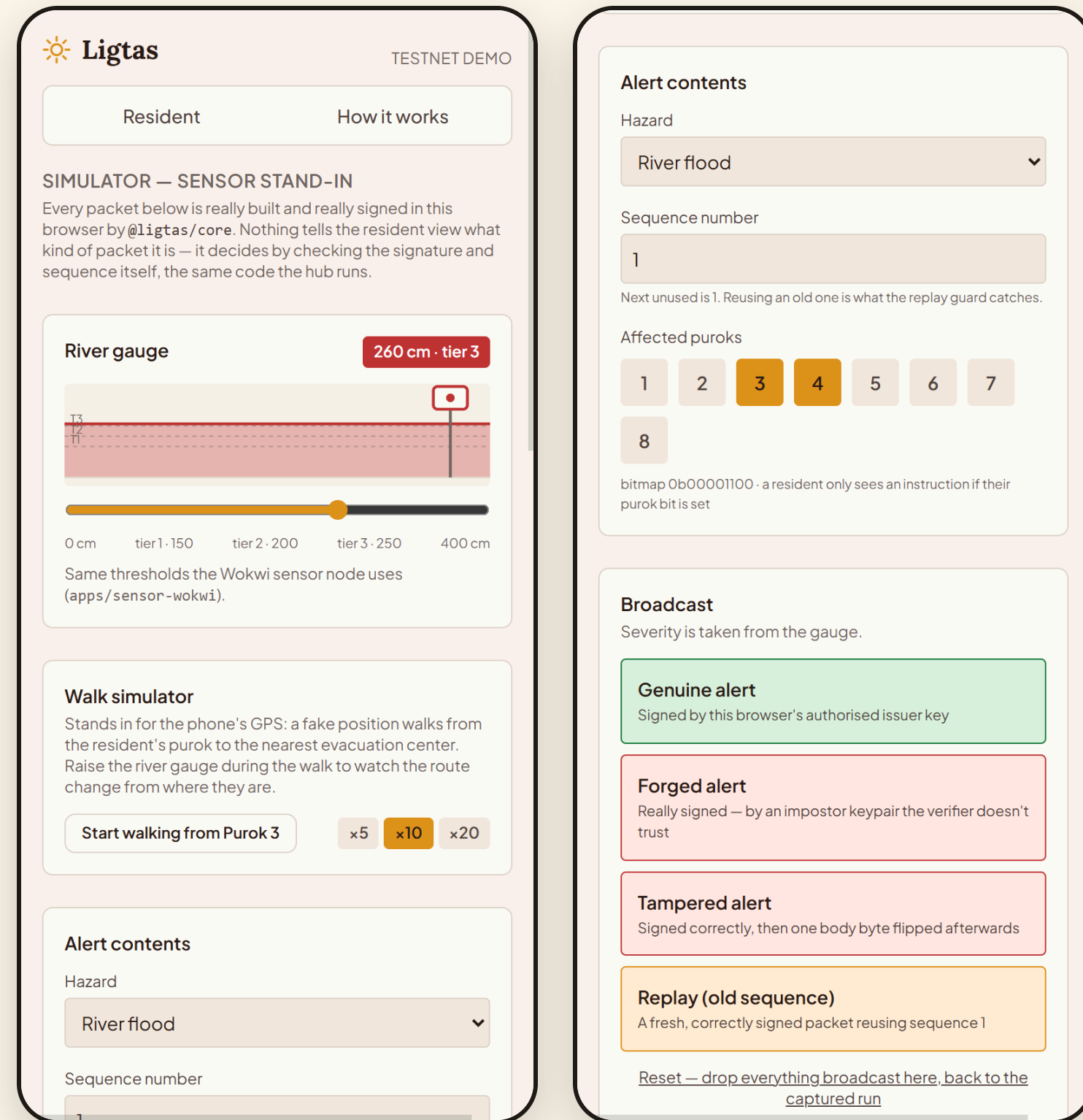
When connectivity returns, the alert hash is anchored and payouts release.



STEP 1 OF 7

A resident opens the app and picks a purok

- One tap to continue. The only thing the app needs is your purok, 1 to 12.
- The first visit caches the app, the map and the latest alerts, so it opens again with no internet.
- The choice is remembered on the phone. Joining a household is optional.



STEP 2 OF 7

The river rises and a signed alert is built

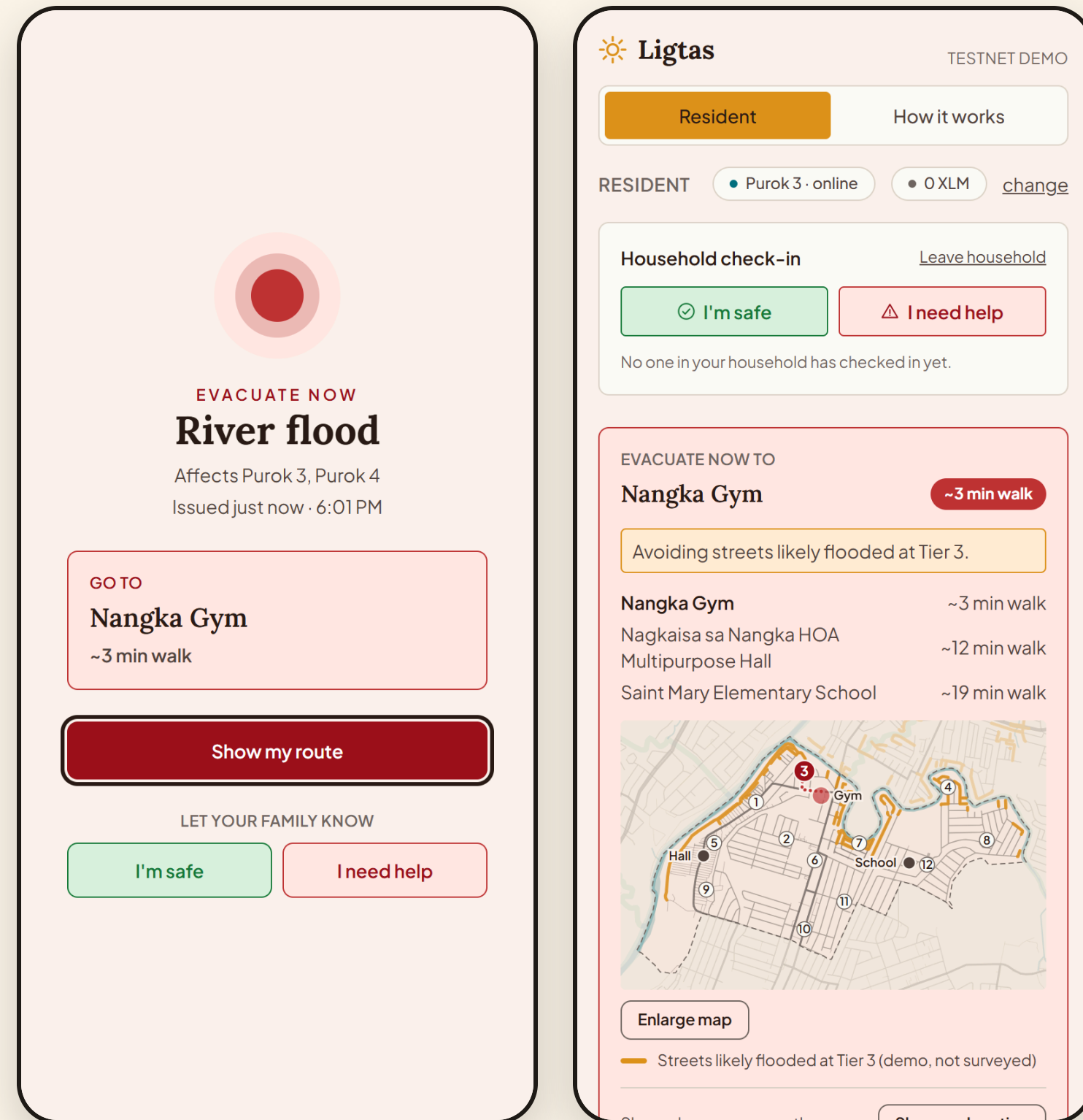
- In the demo, the Tester stands in for the sensor node, with the same 150, 200 and 250 cm tiers as the Wokwi sensor.
- At 260 cm the gauge reads Tier 3, and severity is taken from it.
- Puroks 3 and 4 become one bitmap, 0b00001100. No names, no addresses.
- Genuine alert signs the packet with a real Ed25519 key. The other buttons send the bad cases.



STEP 3 OF 7

Every packet is verified, never trusted

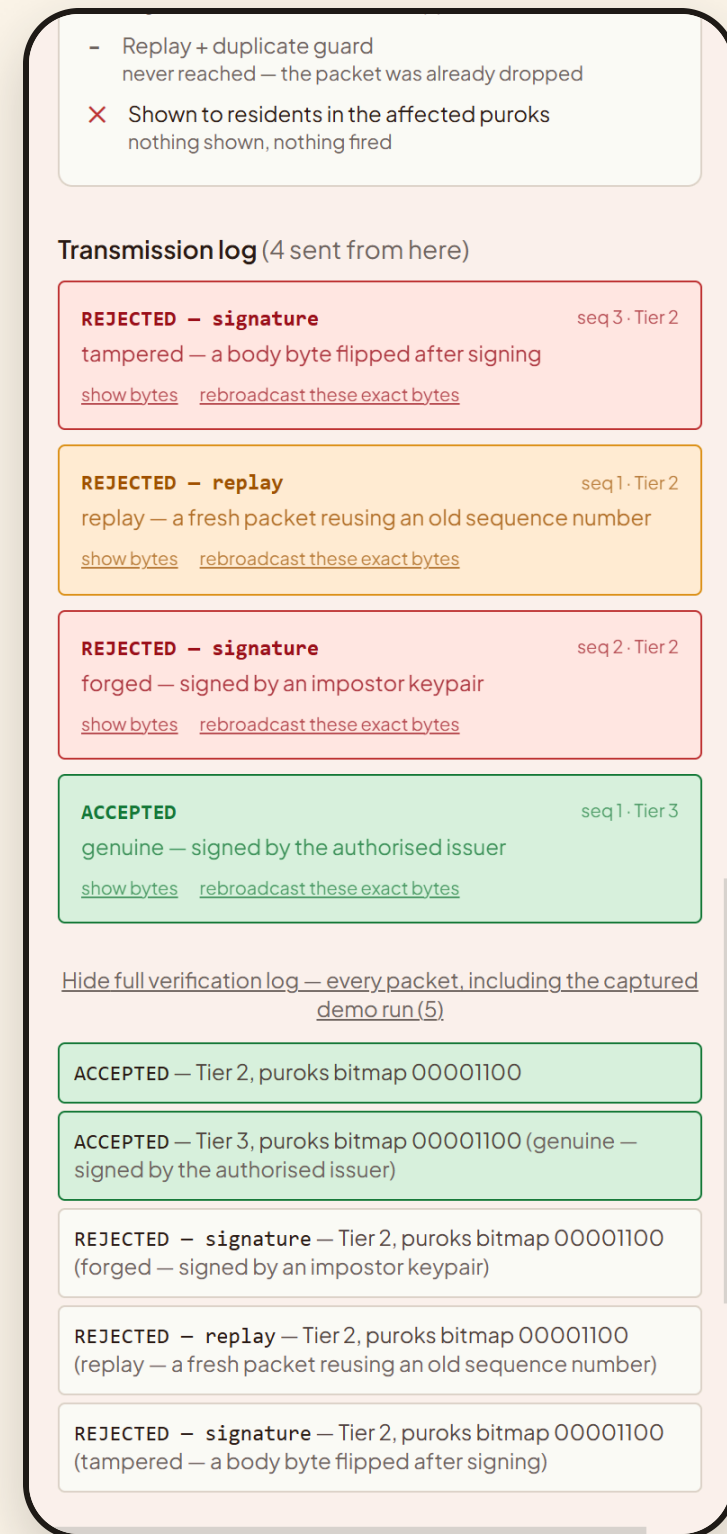
- Left, a genuine alert passes the signature check against the cached issuer list, then the replay and duplicate guard.
- Right, a forged alert fails at the signature check. It is dropped, and nothing is shown or fired.
- The radio hop is not simulated in the browser. It was proven separately against a real Meshtasticator run.



STEP 4 OF 7

The right phone gets the right instruction

- Tier 3 takes over the screen: evacuate now, the nearest center, about 3 minutes' walk.
- Show my route opens the offline map. From Tier 2 up, the route avoids streets near waterways, in orange.
- Only puroks 3 and 4 see this. Purok 1 is told its purok is not affected.

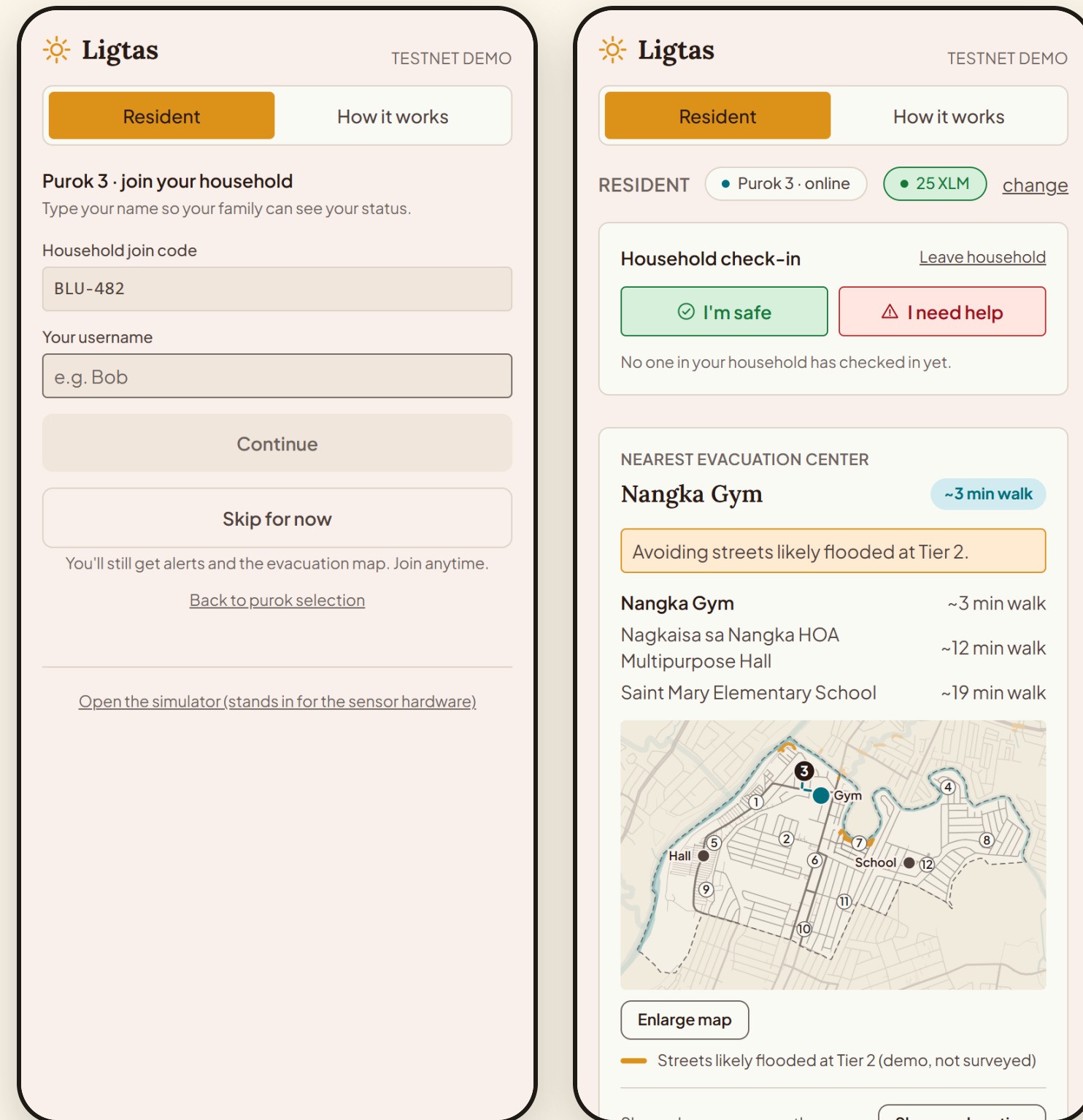


STEP 5 OF 7

Forged, tampered and replayed alerts are rejected

- Genuine: accepted.
- Forged, a valid signature from an unknown key: rejected on signature.
- Tampered, one body byte flipped after signing: rejected on signature.
- Replay, a fresh packet reusing an old sequence: rejected as a replay.

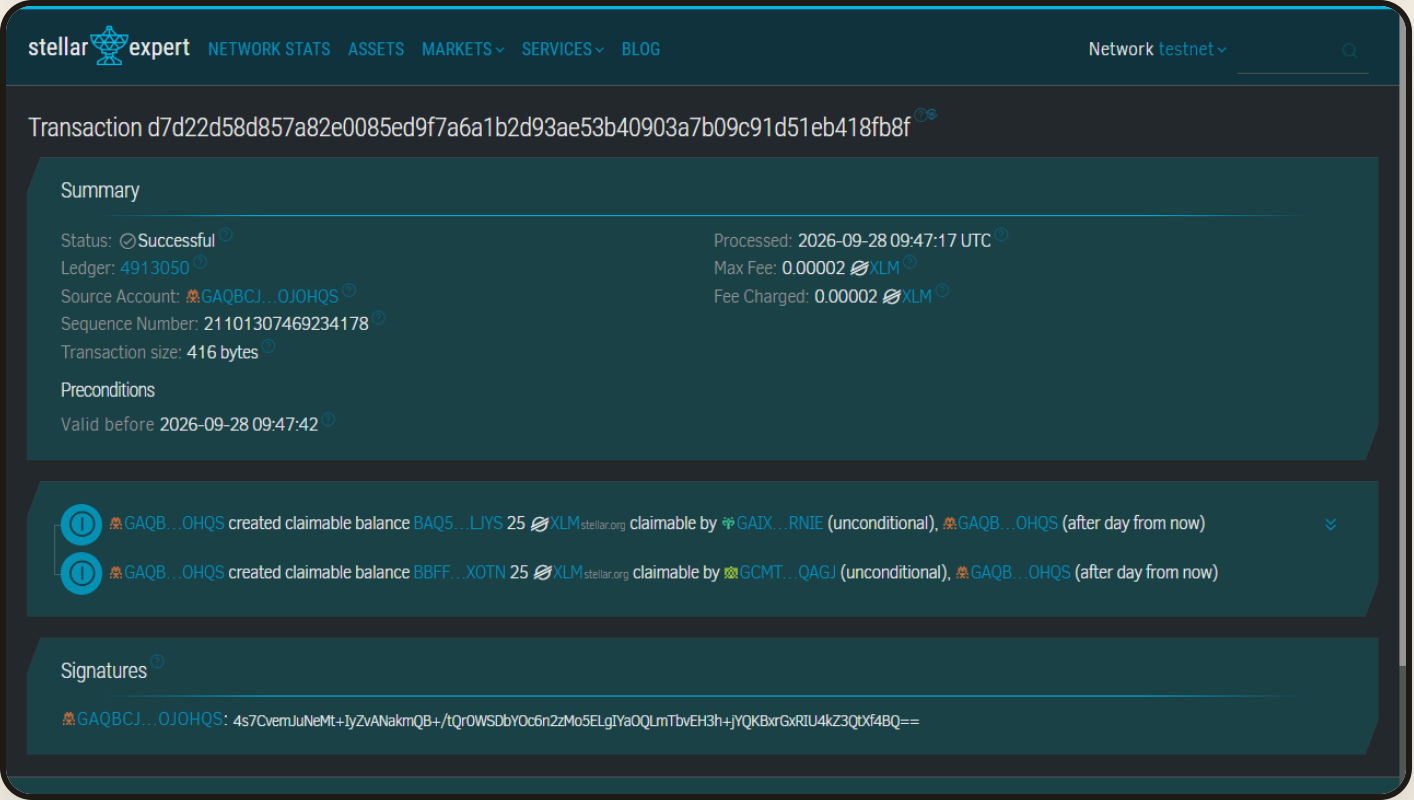
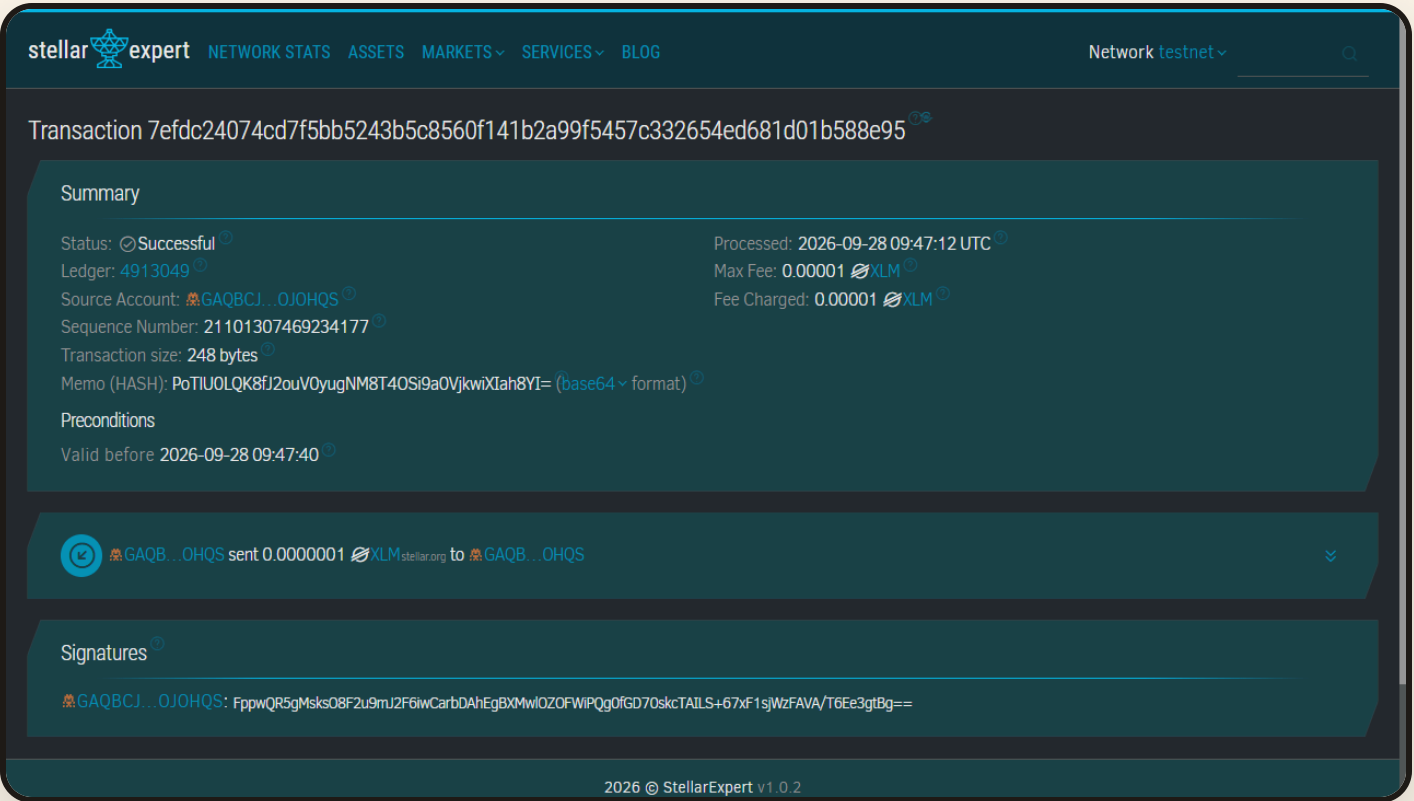
A stock relay forwards any payload unparsed, so only a verifying endpoint can tell them apart. The phone and the hub run the same check.



STEP 6 OF 7

Families check in, and relief shows on the phone

- With a join code, a resident joins their household and taps I'm safe or I need help. Taps are saved on the phone first and sent when a connection allows.
- The green pill reads the household's claimable balance straight from Stellar Testnet: 25 XLM for the Tier 2 alert on screen.
- All optional. Alerts and the map work with just a purok.



STEP 7 OF 7

Connectivity returns: the record and the payout land on Stellar

- The hub drains its queue on its own. The anchor's memo decodes to the alert hash, byte for byte.
- The payout is two claimable balances of 25 XLM, only to the households in purok 3 and 4, reclaimable by the pool after 30 days.
- Both transactions were re-checked on Horizon directly. Stellar Testnet, 28 Sep 2026.

Three layers, each doing one job

The radio

A LoRa mesh carries the warning kilometres across the barangay, with no internet and no phone company.

The signature

Ed25519, verified offline in milliseconds. A cheap radio cannot broadcast a believable fake evacuation order.

The chain

Stellar anchors each alert hash as a neutral receipt and moves the payout. It never delivers the warning.

One identity: the Stellar key that signs an alert is the same key that funds its payout.

What the resident sees

Tier	Level	On the phone
1	Watch	A quiet note on the home screen
2	Prepare	A marigold card on the home screen
3	Evacuate	A full-screen takeover with the nearest center and the way there

Works offline

A PWA that caches on the phone and stays readable out of range.

Offline map

Real OpenStreetMap streets of Nangka, Marikina, with flood-aware routing.

Honest about age

Every screen says how old its information is and warns when it is stale.

Verified live, end to end

Packet sent to a real hub	Result
Genuine, Tier 2, puroks 3 and 4	Accepted
The same packet, resent	Duplicate, a normal rebroadcast
Valid signature from an unknown key	Rejected: signature
Reused sequence, different body	Rejected: replay

Anchor matches

The Stellar memo decodes to the alert hash, byte for byte, re-checked from Horizon.

2 × 25 XLM

Two claimable balances, only to the two households in the alert's puroks.

Stellar Testnet, 28 Sep 2026. The five-node mesh leg was proven in an earlier live run, 9 of 9 checks.

Why Stellar, and nothing more

One identity

A Stellar account is an Ed25519 keypair. The key that verifies a field alert also disburses funds, with no separate PKI.

Memo anchoring

The alert hash goes in a transaction memo: a permanent public receipt, with no smart contract.

Claimable balances

One per matched household, flat by tier at 10, 25 or 50 XLM, reclaimable by the pool after 30 days.

Stellar Classic only: no Rust, no Soroban. Testnet only.

No names or addresses on chain, only account addresses.

About ₱20,000 per barangay

₱20,000

Roughly five nodes at about ₱4,000 each: board, sensors, solar panel, battery, enclosure, siren.

Not one radio per household

At 200+ households that would be ₱300,000. The radio crosses the barangay; the siren and hub WiFi cover the last 50 metres.

Built for ₱0

The mesh runs in Meshtastic's simulator and the sensor node in Wokwi. No hardware was bought.

Money that already exists

Payouts draw on the 5% LDRRM Fund (RA 10121). The household registry is one added column on records barangays already keep.

What this does not prove

Limit	What it means
Simulation cannot measure radio range	Protocol, signatures and settlement are proven. Propagation through real terrain is not.
The map is a demo snapshot	Streets of Nangka are real OpenStreetMap. Purok positions and flood tiers are a demo model, not a survey.
Parametric is imprecise	Payout follows location and threshold, not damage. Speed is bought with precision.
The phone hears only while the app is open	There is no push yet. A phone in a pocket depends on the siren.
The proof is two live legs	The mesh and the hub, app and Stellar chain were each proven live, not as one recording.

NEXT

Two nodes, one real river

- A two-node field pilot, budgeted at ₱4,300, to measure real LoRa range.
- Confirm evacuation centers and purok boundaries with the barangay.
- Push delivery, so a phone in a pocket still hears the alert.
- Real flood survey data in place of the demo flood model.

The end of our pitch. Thank you so much!

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